

NeuroAgent

Building and validating an evidence-driven neurological Agentic System

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Motivation



A new way of testing medical AI

Most research today tests medical AI with **question answering**. Real clinical work is not like that.



Today: question answering

- Multiple-choice exams
- Every clue is already in the question
- One correct option to pick
- Rewards recall, not reasoning

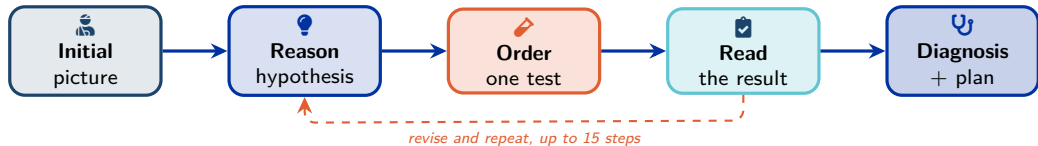


Real clinical practice

- The patient arrives with a complaint, not options
- You decide which investigations to order
- Results come back over time
- Cost, risk and safety all matter

We introduce a **new paradigm**: test the AI on the **whole diagnostic workup**, the way a clinician actually reasons.

Project overview



600 realistic neurology cases across 20 conditions.

🎯 What we measure

- Right test, right order, sensible cost
- Sound reasoning, not memorised facts
- A workup a clinician can follow

💡 Why it is new

- Today's AI exams hand over every clue
- Tests the real skill: what to investigate
- Each case checked by **2+ clinicians**

The agent in action

The goal: a transparent workup, ending in a diagnosis backed by evidence.

The screenshot displays the NeuroAgent interface, which is divided into several panels. On the left, a sidebar contains navigation options: Cases, Dataset, Traces, Rules, and Settings. Below these is a search bar and a list of cases, including 'ALS-H01', 'ALS-H02', 'ALZ-EARLY-H02', 'ALZ-EARLY-H03', 'ALZ-EARLY-H01', and 'FXT1-T00-H02'. The main panel is titled 'Trace Replay / ALZ-EARLY-H01' and shows patient information: '68yo male', 'right-handed', 'Hispanic', 'BM 27.3', 'outpatient', and '97%'. It also displays '135/77 mmHg' for blood pressure. Below this, there are tabs for Overview, History, News Exam, Diagnostics, and Ground Truth. The 'Overview' tab is active, showing a 'CHIEF COMPLAINT' section with the text: 'Subjective memory difficulties and word-finding trouble over the past year'. This is followed by a 'HISTORY OF PRESENT ILLNESS' section containing a detailed paragraph about the patient's symptoms and medical history. On the right side of the interface, there is an 'Agent Timeline' section with a list of reasoning steps. The first step is 'Agent Reasoning' with the text: 'The patient is a 68-year-old male presenting with progressive memory difficulties, word-finding...'. The second step is 'Lab Results' with the text: 'clinical_context: 68-year-old male with non-onset cognitive decline; comprehensive dementia...'. The third step is 'Agent Reasoning' with the text: 'The lab results are informative but not surprising. Reversible causes have been ruled...'. The fourth step is 'Brain MRI' with the text: 'clinical_context: 68-year-old male with progressive memory impairment, word-finding difficulty...'. The fifth step is 'Agent Reasoning' with the text: 'The lab and MRI results now provide a comprehensive picture. Key findings include: 1)...'. The sixth step is 'Drug Interactions' with the text: 'drug: donepezil, current_medications: ["Mefenamic", "Naloxiprine", "Nevastator", "Tamsulosin"]...'. The final step is 'Final Assessment' with the text: 'Primary Diagnosis: Mixed Dementia (Alzheimer Disease + Vascular Cognitive Impairment) (Confidence: 0.85)'. Below this, there is a 'Differential Diagnoses' section with the text: 'Pure Alzheimer Disease - Less likely due to significant vascular burden (Fazlous 2 white matter diseases, lacunar infarct, microbleeds) contributing meaningfully to cognitive profile. Vascular Dementia Alone - Unlikely because hippocampal atrophy (MTA score 1-2) and temporal/parietal cortical volume loss are characteristic of AD pathology; vascular injury alone doesn't explain prominent episodic memory deficit. Psychogenic/Pseudodementia - Ruled out by gradual progressive course over 12 months, objective neuropsychological testing showing true cognitive impairment (MMSE 26/30), preserved awareness of deficits'. Finally, there is a 'Key Evidence' section with the text: 'Clinical Features: Progressive episodic memory loss over 12 months, executive dysfunction (financial mismanagement, driving hesitation), word-finding difficulties—consistent with AD phenotype. Neuropsychological Testing: MMSE 26/30 with specific deficits in delayed recall (epi-logic memory), clock drawing (executive/aparal), borderline category fluency—pattern typical of Alzheimer's. MRI Findings (Crucial): Bilateral hippocampal atrophy (Scheltens MTA 1-2) AND moderate small vessel ischemic disease (Fazlous 2); chronic lacunar infarct in left caudate, scattered microbleeds—demonstrate concurrent AD and vascular'.

Dataset



The dataset



Cases

600

20 conditions



Per condition

30

exactly, fully balanced



Follow-up results

3,800

triggered by actions



How each case is built

- Patient core: history, exam, vitals
- Evidence hidden behind tool calls
- Ground truth: best path, red herrings, unsafe actions



Spanning

- 3 difficulty tiers: straightforward to puzzle
- 396 synthetic + 204 real-report-seeded
- 3 settings: emergency / inpatient / outpatient

*Results are cached in two tiers: **initial** findings when a test is first ordered, and **follow-ups** unlocked by a specific next action.*

How the cases were generated

Two pipelines, one schema: real cases for fidelity, synthetic cases for scale.



Seeded from real case reports

- Source: **MedCaseReasoning**, clinician-written case reports
- Rewritten by **Claude Opus 4.8 + Sonnet 4.6**: findings hidden in tool outputs

204 cases



Fully synthetic, for scale

- Built from per-condition clinical schemas, no source report
- Scale needed to **fine-tune local models**
- Same schema and ground truth as the seeded cases

396 cases

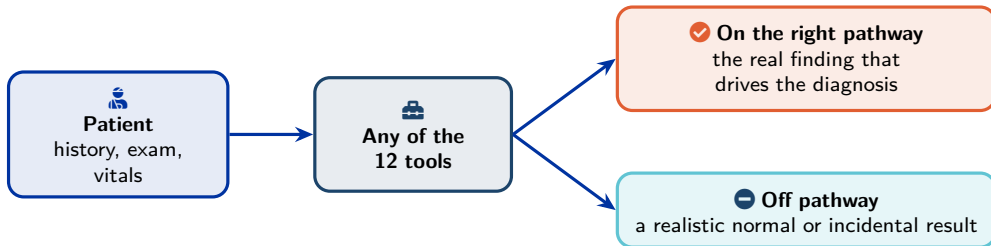
The 12 diagnostic tools

Tool	What it returns
Brain MRI	report with region-tagged findings
EEG	background, epileptiform features, focality
ECG	rhythm, intervals, signs of ischaemia
Labs	panels with reference ranges + abnormality flags
CSF (lumbar puncture)	cells, protein, glucose, NfL, oligoclonal bands
CT scan	non-contrast / contrast, head / spine
Echocardiogram	chambers, valves, ejection fraction, thrombus
Cardiac monitoring	telemetry / Holter / loop recorder
Advanced imaging	MRA / MRV / CTA / PET / DaT-SPECT
Specialised tests	EMG/NCS, biopsy, autoimmune panels
Literature search	guideline and paper snippets
Drug interactions	interaction profile + contraindications

Each returns a structured, typed report with a cost; any tool, any time.

Every patient is a complete picture

Each case pre-computes a **whole patient**, so the agent can order **any** of the 12 tools and always get a realistic answer.



Never a dead end: every tool answers. But each call costs money and a turn, so ordering the wrong test is **penalised**.

A worked example: wake-up ischemic stroke



Turn 1: what the agent sees

- 58 F: woke with slurred speech, clumsy hand
- Last known well 9 h ago: window unclear
- Left face/arm weak, sensory loss, neglect



What the right workup reveals

- CT first: rule out bleed; glucose: no hypoglycemia
- MRI: fresh infarct, deep right white matter
- ECG, echo, carotids, Holter: hunt the cause



The trap, and an off-pathway test

“Chronic, not acute” tempts an early stop, yet the cause hunt is still owed. An off-pathway EEG reads **normal**: wasted.

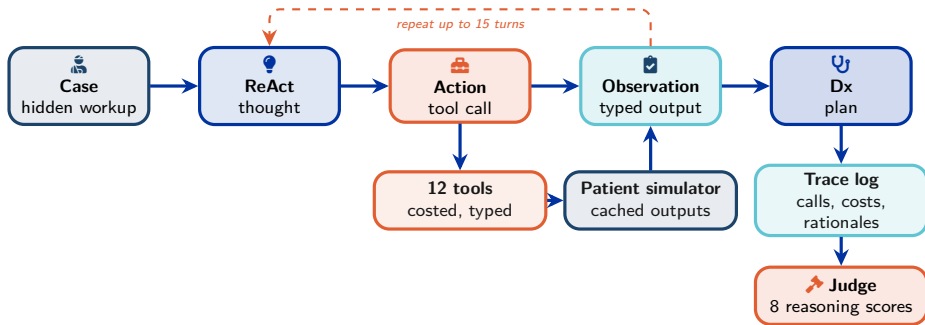


Ground truth: acute ischemic stroke, right corona radiata (ICD-10 I63.541).

Agent



Agent stack: from patient to judged trace



Why this matters clinically

Every run is replayable: ordered tests, revealed evidence, cost/safety events, and an independent reasoning-judge score.

Preliminary base-model results

Model	Composite	Top-1	Top-3	Tool F1	Safety fails
qwen3.5-9b	0.779	35.3%	63.3%	0.56	28/150
qwen3.5-4b	0.671	36.7%	48.7%	0.46	45/150
nemotron-nano-9b-v2	0.565	11.3%	24.0%	0.28	82/150
gemma-4-e2b	0.538	13.3%	33.3%	0.35	72/150
gemma-4-e4b	0.483	12.0%	19.3%	0.26	90/150
nemotron-3-nano-4b	0.339	5.3%	8.0%	0.06	130/150



What the pilot says

50 random v5 cases, 3 repeats per case and model: 900 ReAct traces judged by an independent LLM judge. Qwen3.5 leads on composite score and top-3 calibration.



Failure modes to validate

Lumbar puncture ordered before imaging, unsafe medication advice, text-only fake tool calls, and weak reasoning to positively diagnose functional neurological disorder recur across models.

What those metrics mean

🕒 Rule-based checks

- **Top-1:** final diagnosis matches the ground-truth ICD group
- **Top-3:** correct ICD group appears anywhere in the differential
- **Tool F1:** called tools vs the optimal workup sequence
- **Safety fails:** contraindicated, harmful, or wrong-order actions

🔍 LLM-judge composite

- **Judge dims:** 8 clinical-reasoning scores, each 0–5
- **Composite:** weighted mean, normalised to 0–1
- **Main weights:** diagnosis accuracy + clinical safety
- **Also checks:** evidence use, differential, uncertainty, tool efficiency, red herrings

👤 **Clinician validation:** cases and sampled runs, including field correctness, hypothesis quality, unsafe reasoning, and score/clinical-quality alignment.

ICD = International Classification of Diseases, the WHO standard diagnostic coding system used to check whether the model's diagnosis matches the ground truth.

Fine-tuning approach: SFT w golden trajectories

Supervised Fine Tuning corpus

1. **Input:** clinician-approved optimal action sequence
2. **Trace:** rationale → tool call → cached result → next rationale
3. **Filter:** correct diagnosis, no unsafe / wrong-order actions
4. **Lock:** train only on clinician-validated cases

LoRA / QLoRA

LoRA: adapters; base frozen

QLoRA: 4-bit frozen base

Trainable: 116.4M / 9.07B params (1.28%)

Fits: long ReAct traces on one A100-40 GB

Hybrid model: attention LoRA only on the 8/32 full-attention layers; MLP LoRA on all 32.

Why SFT is not enough

Learns: ideal path

Misses: recovery after a wrong early test

Needs: scored sampled trajectories

RL roadmap: optimise scored trajectories

GRPO (*planned*)

- **Sample:** several workups for one case
- **Score:** benchmark each trajectory
- **Update:** improve vs group average
- **No value model**

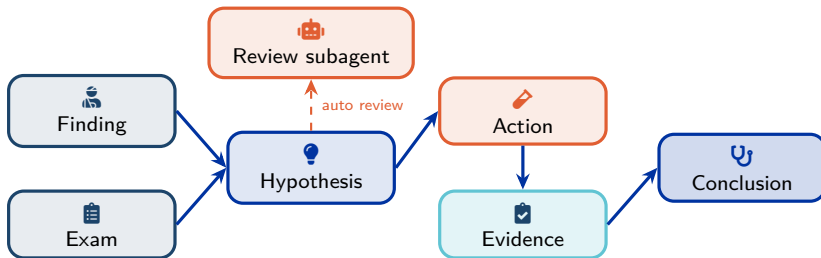
DAPO (*planned*)

- **Target:** longer reasoning traces
- **Keep:** useful hard trajectories
- **Reduce:** noisy clipping effects
- **Focus:** updates where gains remain

Reward signal

trajectory → benchmark scores → composite reward → policy update
diagnosis | evidence | tools/order | cost | contraindications | red herrings | uncertainty

Hypothesis DAG: make reasoning inspectable



Typed nodes

Finding / exam: observed facts

Hypothesis: candidate diagnosis

Action / evidence: test and result

Conclusion: final diagnosis + plan

Typed edges

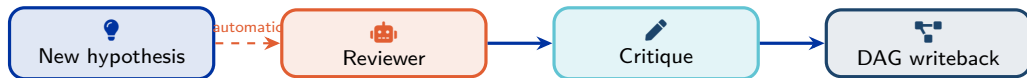
supports: adds weight

contradicts: weakens a hypothesis

triggers: justifies an action

depends-on: records prerequisites

System-triggered hypothesis review subagent



Trigger rule

When: each new hypothesis node

Who: system, not acting agent

Why: critique cannot be skipped

Review checklist

differentials | evidence gaps | contradictions
premature closure | unsafe next actions
test justification

Structured writeback

severity | claim | evidence_node
recommended_fix | clinician_review_needed

Clinician validation

cases + sampled runs
hypothesis quality + unsafe reasoning
clinical usefulness of DAG critiques

Collaboration



Goals for the medical collaboration



1. Validate the cases

- Read each patient as you would in clinic
- Approve, flag, or correct field by field
- Drop cases that are not good enough



2. Validate our scoring

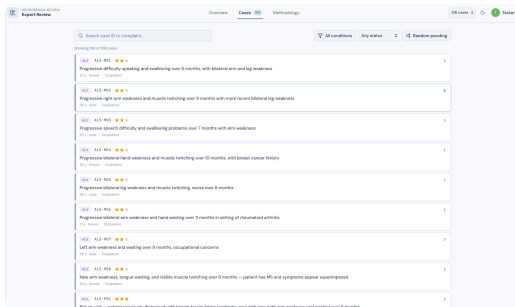
- Score a sample of agent runs yourself
- Compare against the AI reasoning judge
- Confirm scores track clinical quality



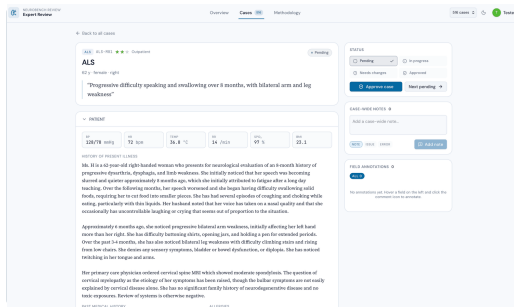
How the review works

- **≥ 2 independent reviewers** per case
- Separate login per reviewer: blind, no shared bias
- Agreement stats $\rightarrow$ fix or drop weak cases $\rightarrow$ higher quality

The annotation platform



 Browse cases: condition, difficulty, status



 One screen per patient: vitals, history, full workup

 review.andreaprotani.com

Reviewing a case

PAST MEDICAL HISTORY

Hypertension, diagnosed 10 years ago, well-controlled on lisinopril

Hypothyroidism, on levothyroxine for 8 years

Osteoporosis, on calcium and vitamin D supplementation

PATIENT > CLINICAL HISTORY
> PAST MEDICAL HISTORY > [2]

Note Issue Error

Describe the issue or note...

⌘+ to save - Esc to dismiss Cancel Add

Alcohol Social 1-2 glasses of wine per week

Living Lives with husband, two adult children

Situation nearby

Functional Previously fully independent; now difficulty

ALLERGIES

Penicillin — rash

MEDICATIONS

Lisinopril · 10 mg · once daily
for Hypertension

Levothyroxine · 75 mcg · once daily in the morning
for Hypothyroidism

Calcium carbonate · 600 mg · twice daily with meals
for Osteoporosis prevention

Cholecalciferol · 1000 IU · once daily
for Vitamin D supplementation

STATUS

☐ Pending ☒ In progress ✓

☐ Needs changes ☐ Approved

Approve case Next pending →

CASE-WIDE NOTES 0

Add a case-wide note...

NOTE ISSUE ERROR Add note

FIELD ANNOTATIONS 1

ALL 1 ISSUE 1

patient.history_present_illness ISSUE

age not coherent with the pathology

🗨 Flag any field: note, issue, or error

☰ Per-case status, every annotation
logged

Approve, request changes, or move on: progress tracked per reviewer.

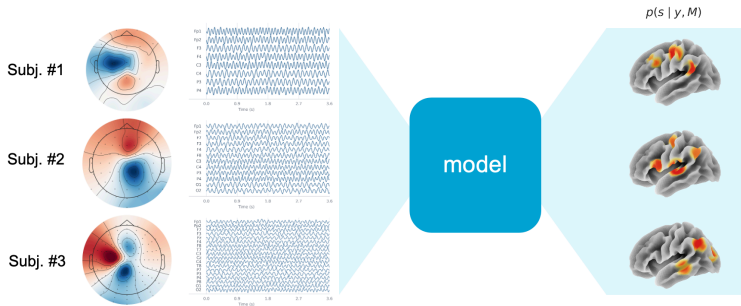
Supervision



Research objective

Goal: EEG source localization → estimate a distribution over plausible cortical sources from EEG, while remaining **montage-agnostic**.

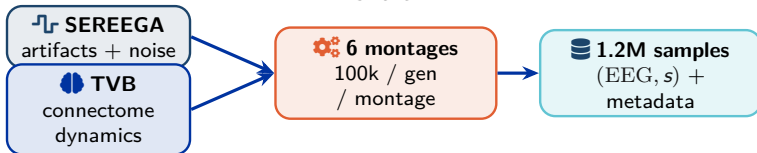
👤 Related MSc thesis project, supervised with **Lorenzo Sibi** | [in linkedin.com/in/lorenzo-sibi](https://www.linkedin.com/in/lorenzo-sibi)



The output should expose **uncertainty**: more than one source map can fit the same EEG.

Synthetic dataset generation

Training needs many paired (EEG, source) samples; real ground truth needs intracerebral stimulation, which is rare.



Artifacts & noise simulated

- Eye-blink / muscle artifacts, line noise, sensor drift
- Pink/white noise; connectivity-driven background (TVB)
-  **SEREEGA** |  **TVB**

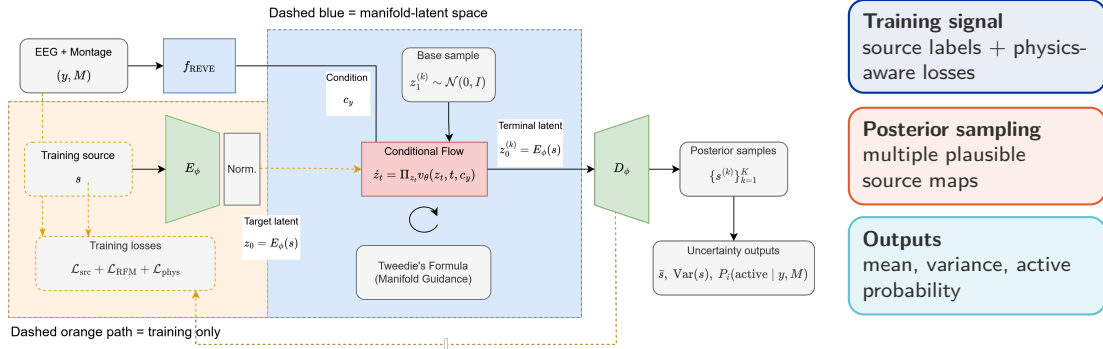


Why synthetic data

- Need large paired sets; ground truth needs real stimulation
- Only **Localize-MI** has it – real-world test set
-  **Mikulan et al., 2020, Sci. Data**

Draft architecture

A conditional flow samples source latents from EEG and montage context, then decodes them into posterior cortical maps.





home.cern